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Value claims of doing Enterprise Architecture

A literature analysis

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Declaration of Authorship

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I hereby declare that I have written this Bachelor Thesis independently, that I have completely specified the utilized sources and resources and that I have definitely marked all parts of the work - including tables, maps and figures - which belong to other works or to the internet, literally or extracted, by referencing the source as borrowed.
I further declare that I have used generative AI tools only as an aid, and that my own intellectual and creative efforts predominate in this work. In the appendix "Overview of Generative AI Tools Used" I have listed all generative AI tools that were used in the creation of this work, and indicated where in the work they were used. If whole passages of text were used without substantial changes, I have indicated the input (prompts) I formulated and the IT application used with its product name and version number/date.

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Abstract

In a world facing increasing organizational complexity and accelerated digital transformation, Enterprise Architecture is frequently cited as a key enabler of strategic change, operational alignment, and IT optimization. However, the claimed benefits of doing EA, understood as the practice of creating a to-be state of Enterprises, vary greatly in different literature.

Following the methodology of a Systematic Literature Review, relevant publications from the industry and academia were selected based on predefined criteria, and benefit claims were categorized according to a framework, distinguishing between IT-related and business-related benefits. The analysis identified a wide variety of value dimensions, ranging from cost reduction and risk mitigation to strategic alignment, innovation enablement, and improved decision-making. These value claims are further analyzed through the lens of EA's roles: informative, instructive, and regulative, revealing a network of interdependencies that jointly contribute to organizational benefit realization.

Beyond categorizing these benefits, the thesis critically examines their measurability and explores contextual drivers of value realization. It highlights notable differences in how academic and industry sources argue for EA's value, as well as existing challenges in tracing and measuring impact. Finally, the work proposes directions for future empirical validation through interview-based studies that can ground theoretical claims in organizational practice. By providing a structured overview of existing value claims, this thesis contributes to a more nuanced understanding of the position and potential of doing Enterprise Architecture in contemporary organizations.

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CHAPTER 1

Introduction

In an era defined by rapid technological change, Enterprise Architecture (EA) has emerged as a critical discipline for guiding organizations through complex transformations [28]. Digital technologies are now ubiquitous, and their adoption is accelerating at unprecedented rates. This early phase of digital transformation is fundamentally reshaping enterprise operations, driven in part by the rise of powerful, accessible artificial intelligence (AI). Beyond mere adoption, enterprises are engaging in a radical re-imagining of digital products and services, as well as intelligent, automated operations. At the same time, growing regulatory demands on digital operations are placing additional pressure on enterprises. Organizations now face the dual challenge of driving shareholder value while satisfying ever-evolving policy and mission goals [28].

Given this multi-factorial challenges and the dynamic nature of current technological progress, many claims are made about how Enterprise Architecture might provide support in those processes as it is said that it cannot only help in tackling challenges but also provides strategical advantage. This bachelor thesis is aimed at providing deeper insights into value claims that are made about enterprise architecture in literature and to explore the different benefit mechanisms. Enterprise Architecture has been defined over 30 years ago [31], and since then a vast variety of organizations and scientists have published papers and materials about implementations, use-cases and benefits. In order to properly incorporate this variety, this paper is based on both academic and industry-sourced papers and focuses on the perception of EA benefits. One the one hand, this duality of sources is supposed to show the variety of claims that exist, on the other hand it is also applied to understand the difference in claimed benefits and overall tonality between industry and academia. Furthermore, this paper dives shortly into the topic of value measurement and value drivers to actually realize those benefits. This is supposed to give a more holistic understanding of the current understanding of EA Value. In the next step, the insights of this work could then be used in an empirical studies consisting of interviews with practitioners. This paper therefore tries to answer the following research question:

- **RQ1:** What are claimed benefits of 'Doing' Enterprise Architecture in Industry and Academic literature?

And the following subquestions:

- **SRQ1:** How do academic and industry papers differentiate in Design, tonality and structure?
- **SQR2:** What are Drivers for EA Benefit Realization according to the literature?
- **SQR3:** What are current challenges and propositions for EA Value Measurement?
- **SQR4:** How can interviews be designed to validate and enrich the identified value claims of doing Enterprise Architecture?

To achieve this, the paper is structured as follows: the first part tackles the theoretical background, diving into the fundamentals of Enterprise Architecture and Value creation. In the second part about methodology, the research design is explained and the selected papers are introduced. The first main-chapter is a deep dive into the identified value propositions and benefit claims, structuring them according to the selected framework. After explaining the benefit claims, the 'Discussion and Implications' chapter 5 is looking at the three sub-questions. To finish this work, we derived implications for interviews based on our research, to guide an empirical study that could validate the existing literature claims.

CHAPTER 2

Theoretical Background

2.1 Doing Enterprise Architecture: Definition and Scope

To reason about claimed benefits of doing Enterprise Architecture means, to have a common understanding of what 'Doing' Enterprise Architecture means. To establish this it has to be said, Enterprise Architecture does not have one clear definition, but rather fragmented interpretations that prioritize different aspects and depth in their definition. From a semantic perspective Enterprise Architecture seems to be a discipline revolving around the architecture, thereby inherent, underlying structure of Enterprises. However, it encapsulates more than just a descriptive discipline. According to the ArchiMate foundation, EA is defined as a set of principles, methods, and models that are used in the design and realization of an Enterprise's organizational structure, business processes, information systems, and infrastructure [20]. Toomas Tamm defines it as the *"definition and representation of a high-level view of an Enterprise's business processes and IT systems, their interrelationships, and the extent to which these processes and systems are shared by different parts of the Enterprise"* [37]. It can be observed that, while definitions differ on a wide range, there seems to be common understanding that EA relates to some kind of structure and relationships in between an Enterprise. From this structural view there seems to be a derived possibility of steering or change, that can be assisted by EA. Bringing those notions together, the book "Enterprise Architecture: Creating Value by Informed Governance" proposes the following definition of EA: *"A coherent set of descriptions, covering a regulations-oriented, design-oriented and patterns-oriented perspective on an Enterprise, which provides indicators and controls that enable the informed governance of the Enterprise's evolution and success"* [18]. This not only refers to a descriptive part but also to an informing part of EA that can steer and drive an Enterprise's development.

This interpretation of EA, where EA accompanies the evolutionary development of an Enterprise, is especially relevant for this paper, as this thesis focuses on 'doing' Enterprise

Architecture. This is understood as the active and purposeful process of creating, modeling and managing architectural representations of an organizations structure, processes and systems in a future state. Rather than only passively documenting, this practice aims to enable strategic decision making, the guiding of transformation and supporting the Business-IT alignment. Although the process of ‘simple documentation’ and ‘purposeful planning and creating’ of EA are not clearly distinguishable, the goal clearly is. Doing Architecture is not about describing an As-Is Situation but rather outlining a To-Be Situation and this paper tries to outline claimed benefits of actively doing Enterprise Architecture in such a transformative process between A and B.

In such a process, EA can take over different roles, that correspond to a purpose or agenda, EA has in its position. According to [11], EA has three important roles:

- **Regulative role:** EA can have the role of actively guiding and prescribing the design of an enterprise. It thereby takes a normative role, that sketches the high-level development of companies and organizations.
- **Instructive role:** The instructive role accompanies the regulative role by providing concrete instructions on the concrete implementation on all abstraction levels. This role and viewpoint covers the Design consequences of the regulative perspective and thereby vision of the enterprise.
- **informative role:** The informative role manifests itself in providing guidance in e.g. decision making. This happens through providing knowledge about the current landscape, application and architectural decisions.

Together, the regulative, instructive, and informative roles show how EA can shape strategy, guide implementation, and provide insight into the current state. Framing EA in this way gives us a simple structure for exploring its many benefits, and will guide the later discussion where benefits can be linked to specific roles of EA [11].

2.2 Enterprise Architecture Frameworks

With a shared understanding of Enterprise Architecture itself, EA frameworks provide guiding principles and tools for developing, representing and managing EA in an organization. These frameworks usually incorporate a structure, concepts, visualizations and a methodology to make EA tangible. While this thesis does not primarily focus on frameworks, they play an important role in shaping the practice of EA and deserve a brief examination. TOGAF, the Zachman Framework, and ArchiMate are among the most widely adopted and influential frameworks and will be briefly introduced [13].

2.2.1 TOGAF - The Open Group Architecture Framework

TOGAF provides a holistic approach to designing, planning, executing and maintaining an organization’s IT Infrastructure and to align it with the overall business goal. TOGAF is organized in four parts and especially the ‘Doing’ EA is a vital part of TOGAF as its centerpiece is the so-called ‘Architecture Development Method’ (ADM) which consists of nine iterative phases to create an Enterprise Architecture for an organization:

1. Preliminary Phase: Defining scope & objectives.
2. Architecture Vision: Creating a high-level target-Architecture of the EA.
3. Business Architecture: Defining business processes, capabilities and structure.
4. Information Systems Architecture: Specifying applications and data to support business.
5. Technology Architecture: Outlining required IT infrastructure
6. Opportunities and Solutions: Identifying improvement areas and potential solutions.
7. Migration Planning: Planning the transition from current to target architecture.
8. Implementation Governance: Overseeing execution and ensuring compliance.
9. Architecture Change Management: Managing and adapting architecture over time.

Next to the Architecture Development Method, TOGAF also offers the Architecture Content Framework, which guides the way the EA-Information is organized. It consists of four components: the content meta-model, the meta-model extensions, content framework artifacts and content framework deliverables. The third part is the Architecture Capability Framework, with the purpose to assist in building and operating enterprise architecture capacity. It comprises four main components: the architecture capability maturity model, the architecture skills framework, the architecture content framework, and the architecture governance Framework [26].
TOGAF offers a comprehensive and structured approach to Enterprise Architecture development and supports organizations in their Business IT-Alignment. The flexibility, scalability and support for governance facilitate application across different industries and organization sizes. As highlighted by Nyale and Karume [26], TOGAF’s strengths lie in its ability to balance structure with flexibility by providing a methodological foundation and practical tools. On the other side, TOGAF is a very broad framework with multiple subjects and domains and thereby also comes with a substantial amount of complexity [26].

2.2.2 Zachman Framework

The Zachman Framework is one of the oldest and most established concepts to Enterprise Architecture which was introduced by John Zachman in the 1980s. Compared to frameworks like TOGAF, Zachman does not define how architecture is developed but rather serves as a structure to ensure the completeness of the architectural representation across different stakeholders and levels of abstraction. The core of the Zachman framework is a six-by-six matrix where the columns represent six fundamental perspectives:

- *Who* - people interacting with the IT systems
- *What* - data and information handled
- *Where* - physical locations and network
- *When* - timing and scheduling aspects
- *Why* - goals and motivations

- *How* - technologies and methods

The rows of the matrix represent different levels of abstraction, from ‘Scope’, which features the reach of the IT infrastructure including mission and goals of the company, to the ‘System Logic’, and at the lowest Level, the ‘Operations Instances’, which covers the actual Instantiation of the ‘who’, the ‘where’, or the ‘how’. The strengths of the Zachman Framework lie especially in its clarity and structure by ensuring that all relevant aspects are considered. It enhances consistency over all the different stakeholder views and supports communication and collaboration by providing a common language, framework and methodology. Nevertheless, the framework itself has a high level of complexity as it requires a thorough and deep understanding of an organization’s operational procedures, information systems and technical foundations and although Zachman offers a Framework for arranging the infrastructure it lacks a clear road map on how to construct and maintain an organization’s IT Systems [26].

2.2.3 ArchiMate Language

The ArchiMate Language was the result of a collaborative research project funded partly by the Dutch government, that had the goal to develop an open and neutral enterprise modeling language to represent Enterprise Architecture. 2009 the ArchiMate Language was adopted as the technical standard from the Open Group to accompany the Open Group’s architecture framework (TOGAF). The ArchiMate Language is an open standard, a tool to model Enterprise Architecture that is supposed to maintain a clear structure while being open for expansion. However, it is more than just a modeling language, the ArchiMate standard consists of several primary components, such as a framework, modeling concepts and an abstract syntax. Its conceptual framework organizes the architecture into layers and aspects, building a similar classification method to the Zachman Framework. However, ArchiMate distinguishes between three layers: the business layer, which is concerned with the products and services offered; the application layer, responsible for supporting the business through application services; and the technology layer, providing the infrastructure needed to run application [21].
ArchiMate emphasizes abstraction aiming to make complex enterprise systems understandable for stakeholders without getting them overwhelmed by details. It enables architecture to visualize relationships within an enterprise in a cohesive manner and its concept of an ‘open standard’ allows flexibility and extensibility to evolve alongside changes in enterprise architecture practices and standards [21].

2.3 Value and Benefits in EA

To contextualize this paper, it is first essential to provide a clear definition of Enterprise Architecture (EA). Only once EA is precisely defined can we critically examine its claimed benefits, and, by extension, its value. Equally important is explaining why understanding these value claims matters, as this rationale underpins the entire analysis that follows.

2.3.1 Understanding 'Value' and 'Value Proposition' in EA

The concept of value plays a central role in understanding the contribution of Enterprise Architecture (EA) to organizations. However, similar to architecture itself, the term 'value' is inherently ambiguous and context-sensitive. This ambiguity leads to diverse interpretations across fields like economics, ethics and management. In the context of EA the concept is often used, but Luis Amaral et al. [29] claim, that it is rarely defined and rather taken as self-evident. This unclear definition however, leads to a lot of ambiguity, especially when it comes to concrete value measurements. Generally speaking, Value is a concept with a full range of semantics and differs in understanding depending on the perspective taken [29]. The Common Ontology of Value and Risk [27] identifies three main interpretations of value: ethical, as in an ethical value guiding the actions of an agent; exchange, as in the value you can exchange something for; and use value, which describes the extent an artifact, process etc. enables an agent to achieve specific goals in a given context. Accordingly, value is not attributed to objects themselves, but rather from the experiences they make possible [27]. As the value perspective of 'use value' is the most accurate description in our case, it can be interpreted as not being an intrinsic property of architectural artifacts or practices but can rather be identified as a situationally ascribed quality that emerges from the alignment between architectural actions and organizational goals.
From a managerial perspective, this has profound implications: doing architecture is not a value-neutral activity. Instead, it is a purposeful activity that can be part of organizational decision-making or re-orientation. The value of doing EA thus lies in its capacity to structure, coordinate, and guide change and operation in a way that enhances overall goal satisfaction across the enterprise.
Understanding value through this ontological lens enables a more nuanced discussion of EA's contribution. It allows researchers and practitioners to move beyond vague assertions of "EA adds value" and towards more concrete, goal- and context-specific articulations of how, when, and for whom this value is realized [27].

2.3.2 Importance of Understanding EA Value for Organizations

Why EA value understanding matters, forms the pillar for the relevance of this paper and according to Jansen et al. [9] this can be broken down into reasons. Firstly, value understanding is essential to access the returns of EA initiatives and to further understand their risks. Secondly, clear benefits can support in aligning different stakeholders and their value expectations.
EA initiatives or EA usage in the process of transformations are often tied to necessary capital in order to enable those capabilities. In order to make a compelling business case you therefore have to guarantee a reasonable return on investment, which makes clear benefits necessary. According to [9] a substantial part of industry practitioners struggle to justify EA budgets and therefore initiatives often face difficulties securing funding due to a lack of perceived value [9]. By articulating the value propositions of EA more clearly, organizations can more effectively assess the potential return on investment and

build stronger, evidence-based arguments for funding EA initiatives.
Enterprise Architecture is also a discipline that involves stakeholders from multiple different backgrounds ranging from Architects, Software Engineers and Business Stakeholders. Therefore, alignment and a shared understanding on Enterprise Architecture is essential. As will be discussed in the following chapters, EA offers multiple dimensions of benefits and addresses different stakeholders. Therefore, it is claimed necessary to create awareness about those values to secure a shared vision and commitment from all different sides: a positive conception of EA and its benefits is essential to align stakeholders and ensure long-term success of EA efforts [9].

CHAPTER 3

Research Design

For this literature analysis, we chose to conduct a Systematic Literature Review steered by the guidelines proposed by Kitchenham and Stuart [15]. This method proposes a framework by planning the review, conducting the review and reporting the review and thereby providing a fundament for an analytical and effective literature analysis. In this report we draw upon these guideline principles to shape our workflow and leverage the framework's emphasis on clear inclusion/exclusion criteria and systematic data extraction, yet allow flexibility in the execution tailored to the research objective of combining sources coming from the industry and academia.

3.1 Search Strategy and Selection Criteria

Literature about Enterprise Architecture is vast and there exists a substantial amount of papers from industry and academia covering potential value claims. Integrating these diverse claims is a crucial step, since academic studies and practitioner-oriented publications each require tailored search strategies and distinct retrieval platforms to capture their respective insights. To further narrow this down, we came up with the following Questions to guide the literature review:

- **Q1:** What categories of value claims regarding 'doing' Enterprise Architecture are identified in the academic and professional literature?
- **Q2:** Which specific value claims are articulated within each of the identified categories?

In this work Q1 provides the foundation of a framework by systematically classifying concrete value claims into organized categories. Using this as a base, Q2 dives into each bucket to identify specific value claims found in the literature.

3.1.1 Literature Search Strategy

This Systematic Literature Review fetches its sources from a selection of search engines, especially targeted at literature and papers coming from academia.

- **Scopus** (www.scopus.com/)
- **Web of Science** (www.webofscience.com/)
- **Springerlink** (www.link.springer.com/)
- **IEEE Xplore** (www.ieee.org/web/publications/xplore/)
- **ScienceDirect** (www.sciencedirect.com/)

These databases were selected as they offer the most important conference papers and journals, that are about Enterprise Architecture. To find the relevant papers and studies, the following search-query was used to search the titles in the databases:

```
("Enterprise Architecture" OR EA OR "IT Governance" OR "IT-Governance
   ") AND (value OR benefit OR strength OR advantage)
```

This search query was based on the two essential components of this work, Enterprise Architecture and Value add. Including IT Governance in the query ensures we capture literature where the practice of Enterprise Architecture is discussed under a governance-oriented context. Many papers use IT Governance and Enterprise Architecture as interchangeable terms and treat them as different sides of the same coin. According to James R. Getter, EA is a foundational component of IT Governance, as IT Governance manifests a continuing activity using the principles and Artifacts of Enterprise Architecture. By querying both terms together, we avoid missing papers that ground EA activities in governance frameworks or, conversely, frame governance initiatives as architectural efforts [7].
Although this paper focuses on the active doing of EA, doing is not included in this search query. By first casting a wide net for any discussion of EA's benefits, we capture both theoretical and practical perspectives—even when authors do not explicitly frame their contributions as doing EA. Only in the following step of filtering the benefits do we identify the claimed values that specifically target doing EA.
To ensure comprehensive coverage of industry-relevant literature, the systematic search will be supplemented by targeted queries for publications from key Enterprise Architecture stakeholders, namely framework developers (e.g., The Open Group), consultancies, and leading IT vendors (e.g., IBM, SAP). In addition to scientific papers, these organizations often maintain their own knowledge repositories consisting of publications, reports or blogs. Therefore, the research design will also include dedicated harvesting of Grey literature via advanced and site-specific searches.

3.1.2 Inclusion and Exclusion Criteria

The literature that is relevant is then also filtered according to the following inclusion and exclusion criteria. Firstly, only english papers are included, guaranteeing that all sources can be uniformly assessed and independently verified. Furthermore, we focus on papers that highlight the benefits and values of doing Enterprise Architecture rather than the advantages of specific frameworks or practices of EA. To validate a certain trustworthiness and scientific relevance, our academic search targets only peer-reviewed papers, studies or books. Since formal peer review is uncommon in industry publications, we include only those white papers, case studies and technical briefs produced by organizations or consultancies with a proven track record in delivering Enterprise Architecture services.

3.1.3 Selection Process

To ensure only relevant studies are included in this review, a multistage pipeline is applied. At each stage, records are filtered by the inclusion and exclusion criteria, becoming increasingly detailed as the process progressed:

1. Search databases and search engines to identify sources based on the search query
2. Except papers based on the inclusion/exclusion criteria
3. Except candidate papers based on the title and abstract
4. Select final papers based on the full text
5. Review and double check selected papers

This pipeline provides the possibility to source from a wide range of papers which are then applied to broad filtering, which gets more granular with every step on the way. By starting with maximum inclusivity and incrementally narrowing the pool, this approach ensures both a broad overview of the Enterprise Architecture literature and a focused, scoped collection of papers that directly address our research questions.

3.2 Result of the academic literature research

In addition to our focused, title-based queries, we also ran a broad full-text search and recorded the total number of hits. This not only highlights how prominent and rapidly growing the field of Enterprise Architecture is, but also underscores the need to carefully refine our search criteria to keep the scope manageable. Consequently, rather than attempting to capture every concrete benefit across all possible use cases, we will concentrate on the more abstract, cross-cutting value propositions.

Search engine results	
Source	**Papers found**
Google	14, 000, 000
Bing	239

Table 3.1: Results from search engines

As shown in the table above, the hit counts differ by roughly six orders of magnitude. While some variation is to be expected, since each search engine relies on distinct indexing and retrieval algorithms, a discrepancy of this scale likely stems from differences in how they parse and process the Boolean AND/OR operators. To gain a clearer picture of the topic's academic relevance, we then carried out the same query in our pre-selected scientific databases.

Full-text results	
Source	**Papers found**
Scopus	123, 365
Web of Science	50, 741
Springer Link	355, 226
IEEE Xplore	90, 375
ScienceDirect	647, 620

Table 3.2: Full-text results from academic databases

Prioritizing a first overview here as well, we applied our query to both Full text and Metadata to understand how present the topic of Enterprise Architecture is. As visible, the number of results varies significantly, which can be explained by the inherent difference in the type of website that we were querying. While Scopus and Web of Science can be understood as multidisciplinary citation databases that aggregate and index content from a wide array of publishers, Springer Link, IEEE Xplore and ScienceDirect are single-publisher platforms that host only their own journals, books and conference proceedings. Consequently, the first one delivers broader coverage (and thus often larger result sets), whereas the latter yield more focused but numerically smaller collections of hits. This also leads to the situation that this search strategy will produce duplicates that are e.g found on Scopus and published by ScienceDirect; this also needs to be taken into account in the filtering process. Furthermore, it became apparent during the research that 'EA' is a heavily used term in medicine and biology contexts, which generated a large number of irrelevant results that stemmed from this term. Because ‘EA’ remains a critical term within the Enterprise Architecture field, we decided to keep it in our search string and will remove unrelated articles in a follow-up filtering step. After the full-text search, the query was used on the titles and yielded the following results.

Title results			
Source	**Papers found**	**Candidates**	**Selected**
Scopus	173	68	11
Web of Science	73	28	2
Springer Link	19	6	0
IEEE Xplore	25	16	3
ScienceDirect	301	6	2

Table 3.3: Title results from academic databases

After carefully reviewing the candidate papers, 12 papers in total were selected to be used in this work. To give a little context to those numbers, and especially the ScienceDirect result, during the search engine querying, it became clear that ScienceDirect was not able to focus only on “IT-Governance” or “IT Governance” but rather replied with all results that contained “Governance”. This led to a relatively extreme ratio of found papers and candidate papers, as 97% of the found papers only included Governance but had no connection to IT Governance whatsoever. Generally speaking, the term Governance didn’t yield as many relevant results as the publication that were hit because of Enterprise Architecture.
Furthermore, the final selection of papers comprises both primary and secondary literature. Although many systematic reviews deliberately exclude secondary sources, we have chosen to include them because our aim is to explore the variety and rationale of value claims, rather than to quantify their frequency or impact. By integrating industry publications with academic studies, we seek to identify what claims exist, understand how they are argued, and uncover any differences between these two source types. Moreover, this approach allows us to examine the broad themes covered by the literature and to highlight the most pressing questions surrounding the value of EA.

No.	Authors	Title
[19]	Matthias Lange, Jan Mendling, and Jan Recker	A Comprehensive EA Benefit Realization Model–An Exploratory Study
[5]	Ralph Foorthuis, Marlies van Steenbergen, Sjaak Brinkkemper, and Wiel Bruls	A theory building study of enterprise architecture practices and benefits
[33]	Graeme G. Shanks, Marianne Gloet, Ida Asadi Someh, Keith Frampton, and Toomas Tamm	Achieving benefits with enterprise architecture
[16]	Sherah Kurnia, Svyatoslav Kotusev, Paul Taylor, and Rod Dilnutt	Artifacts, activities, benefits and blockers: Exploring enterprise architecture practice in depth
[2]	Avsharn Bachoo	Enterprise architecture practices to achieve business value
[18]	Martin Op't Land, Erik Proper, Maarten Waage, Jeroen Cloo, and Claudia Steghuis	Enterprise Architecture: Creating Value by Informed Governance
[36]	Toomas Tamm, Peter B. Seddon, and Graeme G. Shanks.	How enterprise architecture leads to organisational benefits
[6]	Ralph Foorthuis, Marlies van Steenbergen, Nino Mushkudiani, Wiel Bruls, Sjaak Brinkkemper, and Rik Bos.	On course, but not there yet: Enterprise architecture conformance and benefits in systems development
[30]	Luis Silva Rodrigues and Luis Amaral.	Stakeholders perspectives on time horizon and quantification of enterprise architectures benefits/value drivers
[25]	Eetu I. Niemi and Samuli Pekkola	The Benefits of Enterprise Architecture in Organizational Transformation
[31]	Walter F. Santos, Marlos G. Ribeiro, Simone C. Santos, Ivaldir H. de Farias Junior, and Cleyton Mário de Oliveira Rodrigues	The State-of-the-Art of Enterprise Architecture Its Definitions, Contexts, Frameworks, Benefits, and Challenges: A Systematic Mapping of Literature
[14]	Muhammad Baharudin Jusuf and Sherah Kurnia	Understanding the benefits and success factors of enterprise architecture

Table 3.4: Selection of academic papers

3.3 Industry results

Figure 3.1: 2024 Gartner® Magic Quadrant™ for Enterprise Architecture Tools [8]

As the second source of literature, publication from industry practitioners and companies were chosen. As previously discussed, the selection of literature followed a targeted approach aimed at extracting explicit value propositions from leading voices in the field. The Gartner Magic Quadrant for Enterprise Architecture tools was used to identify the most influential technology vendors in the market. From this figure, we chose actors coming from the 'leaders' quadrant - in our case, we used publications from SAP LeanIX, Ardoq, Bizzdesign and BOC Group. Next to those pure EA tools, we also wanted to include consultancies as they are often tightly connected to digital transformations processes. Furthermore, we also used literature about certain frameworks (GEA & The Open Group) and influental thought leaders in the field such as Tom Graves (Real Enterprise-Architecture: Beyond IT to the whole enterprise).
For each selected actor, we looked for papers and publications to identify papers and articles that would be fitting for this thesis. We specifically looked for published surveys or reports, therefore content was only included if it was formally published by the respective organization and explicitly addressed benefits, advantages, or outcomes of doing EA. This approach ensured extracted claims were not primarily marketing slogans, but rather structured arguments or documented case-based insights. Where available, we focused on whitepapers, position papers, and customer case studies that articulated concrete value statements. The resulting compilation, illustrated in Table 3.5, includes the following actors and their seminal publications, which underpin the subsequent analysis of EA value propositions.

No.	Actors	Title
[1]	Ardoq	7 Key Benefits of Enterprise Architecture That Can Change
[17]	Arthur D. Little	Why effectively managing enterprise architecture is key to transformation
[28]	BizzDesign	The evolution of Enterprise Architecture in business transformation
[24]	BOC Group	The Role of Enterprise Architecture in Business Transformation
[4]	Capgemini Invent	How EAM creates business value in different organizational types
[34]	GEA	GEA Enterprise Architecture in Practice: Managing Coherence for Better Performance
[35]	IBM	From Inception to Implementation: Delivering Business Value Through Enterprise Architecture
[22]	LeanIX	The Value of Enterprise Architecture: Why do we need it?
[3]	McKinsey	Enterprise Architecture and Digital Leaders; A Survey Report Based on Research with McKinsey &Company and Henley Business School
[32]	SAP	Explaining the Value of Enterprise Architecture
[23]	Strategy&	Building value through enterprise architecture, A global study
[38]	The Open Group	The TOGAF Standard, Version 9.2
[10]	Tom Graves	Real Enterprise-Architecture: Beyond IT to the whole enterprise

Table 3.5: Selection of industry papers

3.4 Categorization Framework

The next step was constructing a categorization framework, so that value claims could be presented in a understandable manner. The broad perspectives and papers phrased benefits differently, so it was crucial to distill all value claims without losing nuance. To ensure this, framework was supposed classify every point while remaining clear and readable, balancing simplification with comprehensiveness to avoid overly generic categories.
Because this framework will guide later interviews, it must be both systematic and adaptable: high-level value claims form core questions for comparability, while role-specific probes capture stakeholders' unique concerns. In a subsequent validation phase, surveys and interviews can test these claims against real-world practice.
In the existing literature, there are multiple different splits and buckets in categorizing value claims. For instance Shanks et al. [33] distinguish between *project benefits* and *organizational benefits*, highlighting two different abstraction levels. One small scoped project-based, where EA is mentioned in terms of improving IS platforms or improving project management, while the organizational perspective covers high-level strategic benefits like agility or value increases. Another approach to this matter is made by Jusuf et al. [14] which is based on a split into *operational, managerial, strategic, IT infrastructure and organizational.* After carefully inspecting the different ways of working, a framework similar to [18] and [33] was chosen, splitting the value claims in two high-level categories:

- **IT-Related Benefits**
  Encompassing claims about cost and time efficiencies, complexity reduction, governance support, and technical risk mitigation.
- **Business-Related Benefits**
  Covering strategic alignment, organizational agility, innovation enablement, value creation, and other outcome-oriented advantages.

Within each of these two high-level perspectives, we decompose individual value claims into more granular subcategories and then systematically catalog them. This approach both preserves the richness of existing terminology and provides a clear schema for cross-study comparison and interview design. The binary framework inevitably comes with certain limitations. As for instance, cost and time reductions are not exclusively confined to the IT domain but often affect the organization as a whole. Nevertheless, the categorization aims to emphasize where these benefits primarily manifest, whether within IT units or business functions. It is acknowledged that multiple alternative categorizations exist, each with its own merits. This framework does not claim to represent a universal best practice but rather offers a structure that is best suited to the objectives and scope of this study.

CHAPTER 4

Identified Value Proposition

This chapter presents the core findings of the literature analysis by systematically outlining the value claims associated with doing Enterprise Architecture. The categorization follows a dual-perspective framework that distinguishes between IT-related and business-oriented benefits. This separation not only reflects the practical domains in which value is experienced but also aligns with the multifaceted nature of EA's impact across organizational layers.

Building on the theoretical foundation that defines doing EA as an active, purposeful effort to design, manage, and govern enterprise structures, this chapter traces how such activities yield tangible and intangible benefits. The analysis emphasizes EA's contribution in its various roles: informative, instructive, and regulative, across both the IT and business contexts.

The IT perspective focuses on operational benefits including cost reduction, complexity management, risk mitigation, and quality improvements in systems and processes. These values typically emerge from architectural activities within IT departments or transformation projects and are often tied to execution and delivery outcomes. By contrast, the business perspective captures EA's influence at the strategic level, highlighting its role in enabling enterprise-wide transparency, supporting innovation and growth, improving decision-making, and fostering alignment between business and IT.

Each perspective is supported by a structured overview of value dimensions derived from academic and industry sources. These dimensions are illustrated through concrete value claims, which are then examined in more detail in the following sections. The goal is to surface the mechanisms through which doing EA creates value and to make these claims accessible and actionable for both practitioners and scholars.

4.1 IT Perspective

The first bucket of identified values comprises IT-related benefits associated with ‘doing’ enterprise architecture. These value drivers directly affect IT delivery, cost optimization, complexity reduction, and other dimensions of IT operations. Although such IT benefits remain closely connected to broader business outcomes, we isolate them here to highlight advantages experienced specifically by those engaged in enterprise architecture work, advantages that may not be immediately visible outside of IT departments or dedicated EA initiatives.

Cost Reduction	**Academic Papers**	**Industry Papers**
Accelerate delivery & reduce project duration	[5, 19, 14, 6, 36, 30, 14, 2, 18]	[23, 35, 4]
Enhance project management effectiveness	[33]	[34]
Optimize costs and cost-management	[36, 30, 14, 16, 25, 6]	[1, 23, 35, 34, 4, 24, 38, 22, 10]
Reduce maintenance & support costs	[36]	[35, 38, 17, 34, 10]
Improve return on investments	[25, 2]	[35, 23, 38]
Complexity Management		
Reduce organizational complexity	[5, 6]	[23]
Reduce project complexity & overhead	[5, 19, 14, 30, 25, 16]	[23, 3, 32, 4, 24, 38, 17, 10]
Facilitate the integration, of processes & systems	[25, 2, 5, 31, 30, 6]	[32, 28, 38, 10]
Risk Management		
Identify & reduce project risk	[19, 30, 6, 36, 5, 16, 2]	[23, 38]
Reduce IT compliance risk	[18][30]	[1, 32, 28, 23, 10]
Reduce dependency- and change risk	[36, 25]	[3, 35]
Improve security management	[14, 18]	[32, 4, 24, 38]
Quality Increase		
Standardize & drive consistency	[19, 14, 36, 2, 30, 25, 5, 16]	[35, 17, 32, 24]
Enhance IT platform functionality	[33, 31, 6, 30, 14, 25]	[23, 32, 38]
Increase IT asset reuse & mitigate duplication	[16, 2, 14, 25, 18, 36, 5, 6]	[23, 3, 35, 17, 10]
Enhance interoperability	[16, 30, 14, 2, 14, 36]	[38]

4.1.1 Cost Reduction

Based on the literature, cost reduction can be clearly identified as one of the most frequently cited advantage of EA. According to the literature, this reduction and savings are achieved through various drivers. Since this paper is dedicated to the practice of 'doing' enterprise architecture, the savings are largely related to cost reductions in transformation and strategy projects. These projects are often accompanied by IT developments, which are referred to by many sources in the literature.

IT projects can be conceptualized as a dynamic interaction between technical resources and human labor, with labor representing a major cost component; consequently, project duration becomes critically important. Enterprise Architecture (EA) enhances project effectiveness by eliminating redundant tasks and optimizing processes, thereby improving organizational performance and shortening delivery times [14]. Standardization of development work further contributes to time savings by enabling organizations to rapidly undertake platform design and development [5]. Moreover, most value dimensions attributed to EA are expressly aimed at boosting cost and time efficiency through increased transparency and reduced complexity [19].

Regarding overall time efficiency, EA has been shown to accelerate project initiation. According to Foorthuis et al. [5], EA provides detailed models that facilitate up-front project scoping and eliminate redundant development activities. Moreover, by offering a coherent depiction of the enterprise landscape, these models enable critical decisions to be made at the outset and leveraged throughout the project lifecycle.

'Doing' EA to model to achieve a 'To-Be' state is mentioned to also be cost effective in a long term perspective, as maintenance costs can be reduced through classified and reused assets and artifacts [35], this reduction in maintenance and support costs is also often linked, to better reliability and availability of systems [36]. As stated in the tables there are also multiple mentions of improved return on investment and the role of optimizing cost management. On the basis of the mentioned point, EA can be seen as a way to provide guidance and thereby increasing efficiency.

Connecting those points, EA takes multiple roles in the Value Dimension of cost reduction. By enabling effective planning and standardization, EA acts in its instructive role, guiding a project implementation with artifacts such as standardized interfaces or design principles. On the other side, the informative role of EA, enables cost management optimization and management effectiveness by delivering clear, timely insights into systems and processes.

4.1.2 Complexity Management

Complexity Management is also a frequently mentioned point across the literature, firstly. EA is said to reduce organizational complexity as it provides a better insight through a modular presentation of the enterprise, with clear functions [5]. This modularization, also referred to as 'aspect areas' enables a controlled overview of projects and further reduces complexity by standardization of services, processes and systems [6]. EA therefore is said to be able to reduce complexity on an organizational and project level. This value claims seems to be tightly connected to the claim of standardization and rationalization, as mentioned by Lange et al. [19], such overarching optimizations typically result in a streamlined technology and application landscape, reducing the number of platforms and frameworks in use, which in turn lowers overall complexity [16].

However, the reduction in projects is particularly significant, since Bossert and Manwani [3] demonstrate a positive correlation between the urgency of digital transformation and the level of technical debt. In other words, organizations that embark on transformation initiatives frequently confront substantial legacy complexity, which in turn impacts the project's success. According to the paper, in many cases, existing IT landscapes rely heavily on point-to-point interfaces, creating an abundance of direct connections and complex dependencies. Enterprise Architecture can help to mitigate these challenges by structuring processes, information, and applications into coherent building blocks, thereby making the underlying complexity of applications and dependencies visible and more manageable [3]. Moreover, adopting a holistic perspective enables the design of more integrated improvements across both IT and business domains. Niemi and Pekkola [25] note that, in practice, transformation initiatives often unfold in silos because the inherent complexity of organizations makes an enterprise-wide roll-out impractical. EA is said to make enterprise roll-outs not just possible but further facilitates integration of multiple services, by providing a repository of reusable components that contain standardized interfaces. With the connection of visibility over artifacts and services and architectural standards, EA is said to thereby also benefit the integration of new applications [2].

Taken together, these findings show that EA systematically reduces complexity at both the organizational and project levels by providing comprehensive visibility, enforcing standardization, and enabling the rationalization of applications and interfaces. By exposing legacy technical debt and making dependencies explicit, EA further simplifies the landscape, facilitating the integration of new and existing systems and thereby supporting scalable, enterprise-wide transformation initiatives. In this context, Enterprise Architecture again occupies a instructive role in guiding while also performing informative actions and providing an insight into artifacts, applications et cetera.

4.1.3 Risk Management

The reduction of risks for projects but also security topics was also a highly discussed topic in the literature. According to [6] the argument about the reduction of project risk is usually built on the assumption that EA models give a better perspective on risks, as they provide a holistic view on platforms and services. EA is therefore put into a informative role, where it provides an integrated overview to reduce the occurrence of risks [19]. According to [2], project risk is also reduced by helping in project selection and design choices before systems are built, this guidance based on overview is also mentioned in [23], as this insight into different platforms and applications helps in risk mitigation and -identification [5]. Based on [36] EA is also said to help in risk reduction on a program level. As EA can help in better sequencing different projects, and thereby reducing the risk of related projects, as interdependencies can be taken stronger into account in program planning phases.

Regulators are demanding more and more data from organizations and greater control over IT as they try to manage risk, particularly at financial-services companies. At the same time, public service organizations find themselves under greater scrutiny as oversight agencies demand more transparency and internal controls. By providing top-down clarity into IT assets and information, EA can help executives respond to regulatory requirements and external stakeholder demands, offering effective solutions while limiting the cost and distraction [28].

Another value dimension that was mentioned was the reduction of IT Compliance risks. To further understand this point, it is important to clarify that compliance in this context is meant as in regulatory compliance, imposed by governments or agencies. According to [28], Enterprise Architecture enables to integrate Compliance into the planning and execution process of projects by enforcing compliance 'by-design'. Furthermore, digital architectures of solutions can be shared with tools that help in compliance assessment, thereby facilitating the enforcement and control of standards. This becomes increasingly important as standards such as the EU General Data Protection Regulation (EU GDPR) impose highly expensive punishments on Enterprises, if data is not handled properly [32]. Tackling this problem is essential, as with traveling data, there is always risk, according to [1], Enterprise Architecture can be helpful here as it can map data's journey through a company and applications. Practices such as Data Lineage can facilitate the tracking of data travels while also providing clear reasoning for the necessity of the data. By providing this documentation for flowing data, Enterprise Architecture is said to provide the first step for effective IT-Compliance [1].

Risk Reduction in transformation projects is inherently linked to IT Security. EA's primary contribution in this field lies in providing comprehensive insights into system inter-dependencies [24]. However, EA is also said to serve as the logical and organizational framework for embedding and implementing IT security by design [18]. On top of that, EA can also provide an understanding for current needs in IT Security Procurement, such as new solutions etc. Resulting from that, Capgemini Invent emphasizes the importance of raising security awareness among EA stakeholders to ensure effective governance and implementation in their paper [4]. Interestingly, although many papers highlight a wide array of EA benefits, security contributions often receive comparatively little emphasis.

4.1.4 Quality

Closely tied complexity management and cost reduction is also the value claim of standardization. This standardization effects multiple areas of transformation and enterprises. Based on [36] the standardized principles and artifacts can also further reduce the time of initialization. The standardization of e.g. interfaces further facilitates simplification [2] and helps interoperability across the organizational IT landscape [16]. Giving a successful example for this is [36], demonstrating how applications built for the finance department could be leveraged in the Human-Resources department, this shows how EA can help identify those synergies and by providing standardized interfaces, can leverage interoperability.

The standardization also takes place in procurement. Through clear architectural guidance, e.g interface standards, Enterprise Architecture helps narrowing down the field of eligible solutions and thereby also reduces ad-hoc decision often made by individual teams. Through this standardized framework EA is said to accelerate solution selection and lead to a more effective project execution [36].

Next to the standardization, the informative capabilities of EA are heavily mentioned when it comes to IT quality. According to [17], companies often face a heterogeneous IT Landscape, when there is a lack of overview of existing solutions and applications. For 'doing' EA that means, that during a transformation, Enterprise Architecture can help to help identify already existing solutions, and further show solutions that need to be replaced or can be terminated. According to [18] this can drastically reduce delivery time by maximizing reuse of existing solutions. This re-assessment of the current landscape can be interpreted to 'documenting EA', however, the act of 'doing EA' is of course always tightly connected to the assessment of the current situation, as this 'doing' refers to a transformative process between an 'As-Is' and a 'To-Be' situation. That also includes the reviewing of the current artifacts and solutions in order to leverage existing solutions while identifying gaps and redundant systems [36].

Putting these aspects together shows that Enterprise Architecture helps improve solution and project quality by providing clear standards, rationalizing application usage, fostering reuse, and raising awareness of unnecessary duplication. This benefit dimension is thereby closely linked with the dimension of Cost Reduction 4.1.1 as well as Complexity Management 4.1.2 as those dimension are closely tied to high-quality, understandable and standardized applications and processes. As quality of Solutions is tightly connected to the implementation of such, EA is appearing here in inheriting its instructive role of guiding implementation and enforcing regulative strategy through concrete specification.

4.2 Business Perspective

The second bucket of Business Perspective focuses mainly on the regulative and informative role of Enterprise Architecture, where it serves as a strategic enabler for high-level decision-making and organizational planning. Unlike the more instructive, solution-oriented dimensions of the previous chapter 4.1, this perspective emphasizes Enterprise Architecture's capabilities of steering an organization and delivering stretagic insight. In those capacities, Enterprise Architecture operates at the enterprise design level, providing frameworks and insights that support:

- Strategy and Growth
- Decision Making
- Alignment and Governance

The benefit dimensions explored in this section therefore address Enterprise Architecture's value in shaping organizational direction, ensuring strategic coherence, and enabling informed decision-making at the highest levels of the enterprise.

Strategy & Innovation	**Academic Papers**	**Industry Papers**
Provide Transparency & Enterprise Overview	[6, 18, 19, 2, 5, 16]	[23, 3, 32, 24, 38]
Identify change requirements & foster innovation	[25, 14, 30, 19, 2]	[23, 3, 28, 24, 17, 10]
Drive business growth	[14, 25, 30]	[1, 23, 10]
Increase customer centricity & value	[25, 33, 30, 14, 2]	[23, 3, 32, 24]
Decision Making & Agility		
Increase agility & situational awareness	[5, 19, 33, 16, 30, 14, 2, 36, 6, 25]	[23, 3, 32, 24, 38, 17, 22, 10]
Support better investment & solution-selection decisions	[16, 36, 25, 14]	[38]
Provide prioritized guidance	[2, 16, 36, 25, 14]	[1, 32, 24, 17]
Improve decision quality	[36, 25, 30, 33, 18, 31, 14]	[34, 3, 22, 10]
Alignment & Governance		
Align business strategy & IT	[30, 5, 18, 6, 16, 25, 33, 19, 14, 2]	[3, 32, 24, 28, 38, 23, 22, 17, 10]
Foster collaboration & shared knowledge	[16, 25, 6, 19, 2, 30, 31, 14]	[1, 32, 35, 34, 4, 28, 24, 10]
Enable governance standards	[25, 6, 30, 2, 18, 14]	[3, 32, 35, 4, 28, 22]

4.2.1 Strategy and Innovation

According to the literature, Enterprise Architecture can benefit organizational strategy in different ways. The informative role of EA especially comes into action by its way of delivering a holistic multilayered understanding of the enterprise, therefore its, people, processes, roles, goals and policies [18]. This consistent and stable view about the enterprise and its assets, can help in developing the overall strategy of an enterprise [35]. According to [5] EA is said to equip management with this integrated, enterprise-wide perspective that aligns diverse departmental objectives, enabling the development of a unified strategy to maximize overall organizational benefit rather than isolated, locally optimized decisions [5].

From an innovation management point of view, Enterprise Architecture gives leaders a clear picture of how the organization is built and where its complexity lies, making it easier to spot where changes are needed [19]. With this transparency, teams can roll out new ideas faster and understand in advance how those changes will affect the business [2]. When the EA function is well set up and linked to overall strategy, it has near-complete knowledge of the IT landscape and business data. This lets it find gaps, recurring problems and new opportunities for both IT and wider business innovation [17]. By simplifying the IT environment and reducing tight dependencies, EA also lowers the hurdles for adopting emerging technologies like IoT, AI or machine learning. In this way, EA turns IT into a flexible platform that supports experimentation and helps the organization stay competitive.

From a growth standpoint, EA helps organizations turn business strategies into real, implementable technology solutions by mapping where they are now ('as-is'), where they want to go ('to-be') and how to get there [35]. It gives business units clear visibility into the financial impact of their decisions, so they can see investment outcomes, achieve faster returns and boost profitability [2]. Beyond immediate gains, EA also allows teams to model multiple future scenarios without affecting live systems, helping them predict new capabilities, demonstrate proof of value, design strategy and execution plans, and forecast analytics—preparing the organization for rapid changes in technology and processes [1]. Another point related to overall strategy and growth is the enhanced customer centricity, which is mentioned by [25]. According to [33], EA can lead to a better ability of delivering better customer service or experience, as EA can support a deeper knowledge of customers. Furthermore, EA is said to facilitate the translation of customer needs into concrete IT solutions [3].

4.2.2 Decision-making and Agility

From a perspective of agility and decision-making, the literature mentions various benefit dimensions of Enterprise Architecture, mostly revolving around how EA in its informative role provides insights that foster decision making and the ability to change and stay agile.

Agility in this context is meant as the organizational capability to sense changes in their environment and deliver quick and effective reactions. This is supposed to enhance a company's capacity to capture and create value. The concrete capabilities and activities include adapting products and services, supporting new technologies and reacting to external changes in customer behaviors as well as expanding into new markets [33]. Enterprise Architecture's contribution is said to stem from its provision of transparency [19] that is supposed to allow organizations to effectively deal with changes, adjust quickly and build organizational resilience [24]. This claim is also backed by Capgemini that labels Enterprise Architecture as a key enabler for the said agility as well as Innovation [4]. The positive impact in relation to agility is also tied to the flexibility that EA introduces in defining organizational structures and IT platforms [24]. By promoting the decoupling and modularization of Information Systems platforms, EA fosters highly adaptable applications, which makes them easily reconfigured or replaced to meet evolving business needs. This modularity is supposed to allow individual units to tailor solutions locally without undermining enterprise-wide standards. The result is an infrastructure that can absorb new technologies, support changing processes, or integrate acquisitions and divestitures with minimal friction. Such flexibility keeps multiple strategic options open and enables faster, more effective responses to market shifts, giving the organization a clear agility advantage [37].

Next to the Benefit Dimension of Agility, there is a strong notion of Enterprise Architecture as a tool for decision helping in the literature. According to [1] decision-makers often feel overwhelmed with an insufficient data basis to fully understand potential choices for decisions. To stress this even more, according to [1] more and more companies are experimenting with ways of work and agile approaches, where teams get even more decision-making power, leading to an even more drastic need for reliable information. To bridge this gap, Enterprise Architecture is said to be able to provide key information to speed up decisions [33]. By capturing relevant facts, the informative role of Enterprise Architecture can underpin also future decision, e.g. by documenting existing systems or defining principles, EA can thereby support decision making in providing an overview and defining decision processes [36]. Even though this benefit emerges more out of a 'documenting' of Enterprise Architecture rather than a 'doing' perspective, this documental process and understanding of an 'As-is' situation is deeply linked to the designing of a 'To-be' state and thereby the doing-act of Enterprise Architecture. EA teams are also said to act as educators for both IT and business stakeholders, sharing broad knowledge and proven practices to support more informed IT-related decisions [36]. Unlike the targeted guidance on specific applications or technologies captured in EA artifacts, this educational role provides generalizable insights and best practices that stakeholders can apply to analogous contexts and future decision scenarios [36].

Furthermore, EA is also said to play an important role in procurement of application, by offering the opportunity to assess the impact of a new product, but, according to [18], it also benefits decisions by giving insights into whether or not enterprises are able to deliver the product themselves, or which parts of new landscapes are better outsourced. Furthermore, if information about an enterprise is more precise and clearly traceable, it is said to not only deliver higher quality in results but also higher predictability [35]. This act of validating investment decision is called opportunity assessment and incorporates various activities from early-stage project briefs and option papers to detailed conceptual and preliminary solution architectures, to give stakeholders a clear, high-level view of proposed IT initiatives at each approval step [16]. Enabled by Enterprise Architecture, this structured approach enhances transparency of anticipated business benefits, supports robust benefit-to-cost analyses, and helps identify and mitigate implementation risks, thereby improving the overall quality of project delivery [16]. Another activity next to opportunity assessment is business capability modeling, which is performed in the field of Enterprise Architecture. Associated with this process, Kurnia [16] points out that business capability modeling clarifies organizational priorities by enabling executives and architects to align on strategic capabilities, assess their criticality and required maturity levels, and define targeted IT investment programs to advance maturity.

4.2.3 Alignment and Governance

The last bucket of benefits, and one of the most frequently mentioned value dimensions is the improved alignment and governance. According to [36] this factor is actually under the top reasons why companies even invest in EA. In this context, alignment describes the role of an IS platform in supporting business needs. It can be said that a platform is aligned if it is linked to and supports an organization's strategic and operational objectives [36]. Effective business-IT alignment is paramount in transformation processes, the very activities encompassed by 'doing' enterprise architecture, because these initiatives inherently span both IT and business domains and are often driven by digitization goals. According to Strategy& [3], in organizations that do not have IT in the heart of their business core, IT has been historically positioned as a support function. Resulting from that, IT services also mainly took the role of utilities rather than being understood as a strategic instrument. Resulting from that, IT-departments are said to tend to be excluded from strategic level discussion [3].

Enterprise Architecture is said to improve this imbalance [14]. Based on the paper "How enterprise architecture leads to organizational benefits" [36], it is described that EA adopts an organization-wide perspective on business requirements and the requisite digitized processes, thereby enabling information systems to effectively support those needs [36]. Successful alignment through Enterprise Architecture is said to improve an organization's ability to identify and correct misalignment between individual projects and overarching strategic outcomes at the earliest possible stage, thereby preventing wasted effort and resources. Furthermore, it is supposed to ensure that data and information management initiatives are directly tied to core business objectives so that every information asset delivers measurable value. It fosters the creation and maintenance of a unified vision of the future that is embraced equally by both business and IT communities, ensuring that all stakeholders share common goals and expectations and it is further said to underpin effective, integrated change planning that rigorously maintains coherence between business processes and supporting technologies throughout the life-cycle of any transformation effort [18]. This improved alignment and the enhanced communication is said to facilitate business goals [19].

Beyond strategic alignment, Enterprise Architecture also empowers effective communication and collaboration to bring strategies to life. One of the mentioned activities in the literature is e.g. the capability mapping which acts as a common language to form a bridge between IT and Business. Furthermore, Enterprise Architects are said to play a proactive role in identifying and engaging key business and IT actors [4], establishing trusted relationships, and coordinating their efforts to maintain coherence across initiatives [16]. Through consulting and mentoring, EA practitioners are said to educate project teams and business leaders on technology trade-offs and architectural principles, fostering a holistic mindset that harmonizes planning decisions [16]. This transparency and shared understanding are not only supposed to smooth internal collaboration but also to ease integration with external partners and ecosystems [19]. Finally, by embedding stakeholders in evidence-based decision processes and anchoring choices in EA principles, organizations might cultivate an inclusive dialogue that surfaces conflicts early, aligns priorities, and accelerates agile responses to change [36].

The governance aspect of EA can be identified as an underlying aspect throughout many different value dimensions, and also lies on on the bottom of many dimensions. The improved governance is said to be observed in controlling and reducing costs often connected to the standardization and clear overview. The governance of technological solution is also frequently connected to improved technical interoperability across the organizations IT Landscape [16]. Therefore, alignment can be considered one of the more abstract value dimensions of Enterprise Architecture (EA), as its concrete business benefits often do not materialize immediately. This value dimension is relevant across all three core roles of EA. From a regulative perspective, EA enhances alignment by establishing a holistic strategy and overarching organizational governance that integrates various stakeholders and business units. The instructive role of EA contributes by ensuring coherent implementation along the entire value chain, upholding governance standards, and providing guidance that facilitates translation between diverse business domains. Finally, the informative role of EA supports alignment by creating a shared informational foundation, enabling stakeholders to work toward common goals based on a unified understanding of the enterprise landscape.

CHAPTER 5

Discussion and Implications

After presenting the results of the research and engaging in the different value drivers, this chapter is aimed to engage in a discussion of what those claims mean for both theory and practice and how they can be interpreted. Through doing this, not only the literature itself but also this work can be questioned and discussed. Starting with the benefit framework and the corresponding sub-categories, the goal is to understand dependencies and the connection of the roles of EA. Secondly, this chapter contrasts industry and academic perspectives by examining how motivations, terminology and emphasis differ between practitioner-oriented sources and scholarly literature, and what these distinctions reveal about each community's core priorities. Thirdly, this chapter explores the realization and measurement of said values by giving an reviewing the current literature regarding the measurement and realization. It discusses how values can be realized and connects them with the maturity concepts of EA, furthermore, it gives insight into the temporal factor of benefit realization and how successful Enterprise Architecture can be fostered and facilitated. Furthermore, we also shortly discuss the topic of measuring EA impact and quantifying concepts to track value creation as well as diving into the topic on how to empirically validate the findings by conducting interviews.

5.1 Discussion of Benefit Dimensions

The first main chapter was targeted to capture value dimensions of 'Doing' Enterprise Architecture, and independent buckets were developed to organize the wide range of claims. This distinction between different perspectives (IT Perspective and Business Perspective) has the goal of providing a framework to make dimensions better understandable. During the research and collection of claims it became apparent, that practically a clear separation is neither in the high-level categories nor in the sub-categories possible as they are so deeply intertwined. From the literature it became visible, that EA does not offer parallel benefits, but rather a network of values that correlate and improve each other.

For example, the clear, modular structures that enable an organization to scale efficiently also create the freedom to experiment, fueling innovation. That innovation, in turn, generates new products and services that drive revenue growth and deepen customer engagement. To give another example, transparency in architectural artifacts not only enhances decision quality and cost optimization but also clarifies system interrelations and reduces maintenance overhead. Standardization likewise lowers complexity and raises the level of abstraction in models, making a holistic enterprise overview possible. That overview, in turn, fosters the organization's capacity to manage change and risk and strengthens both governance and strategic alignment. Furthermore standardization has impacts on cost-effectiveness as well as on maintenance, interoperability or quality. Therefore, borders between the single value dimensions tend to merge and are the subcategories are most definitely hard to translate and distinguish in a practical situations. However, this means, this has also an impact on the roles of Enterprise Architecture. As discussed in the background chapter 2.1 EA can take different roles, from regulative to instructive to informative. Following the observation of our concrete value dimensions, it can be deducted, that the value of Enterprise Architecture arises from an interplay of the different roles, that also seamlessly merge into each other. As prescriptive (regulative) role builds on a clear understanding of a companies artifacts and systems (informative) and unfolds its true value in translating abstract guidance into clear principles (instructive). Likewise, the informative role of EA consists also of an abstraction, that is ideally tailored to the high-level business needs, thereby presenting information in a way that fosters and helps in developing regulations. In practice, every performance improvement, whether faster project delivery, reduced risk or greater agility, emerges from a loop in which information shapes principles, principles guide instructions, and the results of those instructions generate new information. Thus, rather than operating in isolation, the three EA roles form a tightly woven system whose combined effect unlocks the full spectrum of EA benefits.

Consequently, efforts directed at one dimension routinely generate secondary benefits in others, indicating that EA value propositions cannot be understood in isolation but must be viewed as an integrated system of mutually reinforcing effects. This offers positive synergies, however, it also makes it more complicated to track business value to specific characteristics or roles of Enterprise Architecture.

5.2 Comparing Academic and Industry EA Literature

5.2.1 General Layout and Structure

As this paper examined industry and academic papers, there were also observable differences in the way those papers were structured, formulated and presented. Driven through different ways of working, motives and target audience, they significantly differ while also sharing some connections.

The Academic sources are explicitly peer-reviewed papers, journal articles, and other formal documents that emphasize comparability and adherence to rigorous scholarly standards. In contrast, industry sources encompass books, consultancy reports, and company websites. While academic works are typically accessed via specialized databases and may not be freely available, most selected industry materials are directly accessible. Notable exceptions include proprietary research consultancy reports from Gartner and Forrester, which often require subscriptions or purchase, however, those sources were not included. All together, industry sources were on average short and more diverse in their presentation, while academic papers were primarily papers that were more detailed and longer.

5.2.2 Methodology

From a content perspective, the greater length of academic papers is largely due to their extensive exposition of background and methodology, whereas industry papers tend to dive straight into core content with minimal preamble. Scientific articles typically begin by surveying the literature, explicating theoretical frameworks, and justifying methodological choices, so that the actual findings occupy only a small portion of the text. This structural difference also affects readability and intended audience.
Interestingly, publications from consultancies tend to be more extensive and adopt a more methodical style than other industry outputs. For example, the survey report "Enterprise Architecture and Digital Leaders" by McKinsey & Company and the Henley Business School [3] and the Capgemini Invent study "How EAM Creates Business Value in Different Organizational Types" [4] display layouts and methodological rigor that closely resemble academic papers. By contrast, shorter pieces such as "7 Key Benefits of Enterprise Architecture That Can Change Your Organization" by Ardoq [1] are concise and focus immediately on practical insights without broader introductions or methodological exposition. Similarly, publications by LeanIX [22], presented in a wiki-style format, and the source [32], available on the SAP Learning Platform, likewise prioritize concise, practical guidance over detailed methodological exposition.
There is also a huge difference in the methodology itself, as academic papers use existing research and papers as a base, and their content is backed by citations, and clearly documented sources, resulting in long lists of references. Contrary to that, most of the industry papers do not use external sources for their claims or use no sources at all, this is especially true for [28, 32] or [1] that list no sources at all. Most of the papers use internal sources or surveys, complicating objective verifiability.
It can also be observed, that most of the sources from academia and industry use interviews-concepts to validate benefit claims. For example This combination of claims and empirical data collection can be observed in [6, 36, 25, 14] of the academic papers. From an industry perspective, empirical data is made use of e.g. in [3, 23] or [17]. However, it can be observed that the respective subjects that are the sources vary highly between academic and industry sources. Academic papers use mostly individual subjects from all different seniority levels and practice areas, an example would be [6] where interviewees occupy roles ranging from Managers, to Software Architects and Maintenance Engineers. This broad coverage of roles is exemplary but can observed in the majority of interview-supported papers such as [25, 14]. Contrary to that, the individual subjects

in the industry papers are for the most part top-level executives, such as CIO's, COO's etc. [35, 23] and the papers refer to the interviewees less as subjects but more as instances representing their companies, such as in [3, 17] or [4], the focus here seems to be not the individual benefits, but rather the organizational benefits as a whole.

5.2.3 Content and Focus

Another interesting difference is the amount of benefits and the difference in detail between industry and academic sources. While academic papers often cite a wide variety of benefit claims, exemplary [30] with 29 value dimensions or [14] with 37 value dimensions, the numbers are observably higher than the numbers in non-academic papers. Industry papers often focus on fewer key areas, on average 3-5 claims, this also expresses a need for clear and precise communication of industry papers in order to persuade their clients. Interestingly, most of the industry sources introduce the benefits of EA based on problem statements and pain-points. While academic papers often start with a scientific need for papers, meaning whitespace in the current literature, industry sources frequently use economic and entrepreneurial challenges to argue for EA as a tool. Therefore, the benefits of EA are often more presented as a solution to a problem, than an independent value creator. As an example, the very first sentence of "The evolution of Enterprise Architecture in business transformation" [28] is "Organizations of all sizes and industries face an increasingly complex and rapidly changing business environment that may hinder successful business transformation." this emphasizes the need of industry sources to make a business case, as for a lot of the sources, there is also a product to be sold that stands behind the articles and publications, reflecting different goals: scholarly theory-building vs. vendor-oriented persuasion.
To communicate the different findings, both types of resources use graphical presentations such as bar-charts or tables to underpin statistics, model value chains or present relationships between dimensions of benefits and enterprises such as in [3].

5.2.4 Tone and Argumentation Style

As already mentioned above, industry papers seem to use a more approachable style and use rhetorical methods to underpin their points and examples. For example, one industry article frames an EA challenge with a vivid metaphor: "Every organization acquires a pile of old applications that sit in the metaphorical corner collecting dust." [1] This illustrates how industry literature favors approachable, metaphorical language to communicate key points immediately.
Beyond style, the two genres also differ in their rhetorical stance: academic papers usually adopt a descriptive tone, presenting statements as conditional observations grounded in evidence, while industry sources often use a more prescriptive, action-oriented tone. For instance, Lange et al. observe that *"A common understanding of EA should be established for both business and IT employees"* [19]. By contrast, the consultancy Strategy& commands its readers to *"Promote EA across the organisation"* [23], not only making the statement shorter but also framing it as an imperative call to action. This active phrasing

underscores the industry literature's tendency toward absolutism, presenting assertions as incontrovertible facts and seldom allowing room for uncertainty. By contrast, academic sources usually put their claims in a clear context, note any limitations or assumptions, and treat their findings as provisional rather than presenting them as absolute truths. They often specify exactly when and in which contexts their results apply and admit that more research might change or refine those conclusions.
Generally speaking, the line of reasoning in industry literature is often based on an approach that can be more understood as a 'case-study' rather than systematic interviews or other methods, therefore, providing more anecdotic evidence and support. Those papers often make use of single examples to underpin arguments, the paper "Building Value through enterprise architecture" by Strategy& can be seen as one of the more extreme examples where textual parts are frequently accompanied by a quote from people obtaining top-level roles, that embrace the content. This style can almost be referred to as 'argumentum ad verecundiam' therefore a 'appeal to authority', where single, high-authoritative individuals are used to support an argument. By invoking the prestige of a CIO or COO rather than presenting systematic data, the author bypasses deeper empirical examination and encourages readers to accept claims on the basis of status alone. While such endorsements can be highly persuasive in a business context, driving rapid stakeholder buy-in, they sacrifice analytical rigor and may mask underlying assumptions or limitations.
This approach can be potentially also explained by the target audience of the industry reports which can be identified as subjects obtaining management roles that can be persuaded to trust the abilities of EA. Sources such as [24] explicitly address business decision-makers by using formulations like *"It forms the basis for your strategic business planning and the development of the supporting information technology."* [24] emphasizing the planning role of the audience subject. Similar tendencies can also be observed in the papers created by the Management (Strategy) Consultancies such as Capgemini Invent, McKinsey, Arthur D. Little or Strategy&. It could be argued that in order to pursue such high-impact individuals, it is an established strategy to use other people with similar seniority and responsibility as an example.

5.3 Drivers of Benefits Realization

Although this study is primarily devoted to examining the claimed benefits of Enterprise Architecture, EA is fundamentally a practical discipline rather than a purely theoretical construct; accordingly, this work also seeks to provide a preliminary overview of the key factors that enable those benefits. To structure those factors, a similar framework to the one in "Understanding the Benefits and Success Factors of Enterprise Architecture" [14] was used, as the paper delivers both deep insights and a rigorous methodology. The paper distinguishes four key dimensions of Enterprise Architecture value:

- **Product Quality**, which pertains to the attributes and integrity of EA artifacts themselves.

- **Infrastructure Quality**, encompassing the formal conditions and foundational systems that enable EA to operate effectively.
- **Service Quality**, which focuses on the actual services and support EA provides to its stakeholders.
- **Organizational Embedding**, referring to the degree and manner in which EA is integrated into the organization's structures, processes, and culture.

5.3.1 Product Quality

High-quality EA artefacts form the cornerstone of realizing tangible benefits. First, accurate representations of the current ('As-is') state is said to ensure that decision-makers have reliable baselines for driving transformation initiatives, while coherent models of the future ('To-be') architecture guide long-term planning and investment [19]. Furthermore, clear roadmaps depicting intermediate milestones enhance stakeholder alignment and facilitate incremental value delivery [19]. Beyond these graphical outputs, the rigor of EA conceptual and strategic grounding plays a critical role: frameworks that are underpinned by sound business rationale and methodological consistency yield products that stakeholders trust and adopt [25]. In sum, it is said to be the combination of precise current-state insights, well-defined future visions, and strategically grounded artifacts that form the 'product' driver of EA benefits [14].

5.3.2 Infrastructure Quality

It is suggested that EA success is further driven by a clear goal for the implementation, ensuring that both business and IT stakeholders share a common understanding of its purpose [14]. Robust governance mechanisms and a supportive infrastructure are also argued to underpin the reliable delivery of EA services. Regular monitoring of compliance with EA standards through conformance assessments, coupled with the use of standardized documentation templates, is claimed to ensure consistency and traceability across projects [5, 39]. Incentive structures, including sanctions for non-compliance, are further said to reinforce adherence to architectural guidelines [5, 39]. Clear governance structures with defined decision-making accountabilities are also proposed to align EA objectives with organizational priorities [19, 4].
Moreover, technical infrastructure and tool support, ranging from repository platforms to modeling environments, are suggested to facilitate efficient EA service delivery and collaboration [14, 19, 4]. Capgemini Invent further claims that clear EA principles and objectives guide tool configuration and usage, while a competent EA team leverages these tools to deliver value [4]. Adequate budgetary allocations and resource provisioning are also highlighted as necessary to ensure EA initiatives are sufficiently funded to achieve their goals [4]. Finally, the maturity of the technical environment, characterized by a well-organized IT landscape and continuous platform improvements, is argued to amplify EA's impact [12, 25].

5.3.3 Service Delivery Quality

Another important aspects is the way in which EA services are delivered, which is said to critically influences stakeholder satisfaction and benefit realization. Effective knowledge exchange, both among architects and between architects and business stakeholders, is said to ensure that architectural insights are understood and applied appropriately [5, 39]. Initiatives such as Project Start Architectures embed EA guidance early in project lifecycles, steering implementations toward architectural compliance [5, 39].
According to [3] and [19] Communication capabilities that educate and engage stakeholders amplify the perceived value of EA services. Compliance validation processes and decision-support mechanisms are also cited to aid management in making informed trade-offs and [19] also states that Integration of EA with actual project implementations ensures that architectural recommendations translate into real-world changes, while stakeholder-focused service offerings maintain relevance to evolving business needs [25]. Conformance assistance, including tailored guidance and tool-based validation, further strengthens trust in EA support services [39]. Finally, establishing metrics to measure meaningful success and extending EA practices across the extended enterprise supposedly help sustain and grow EA benefits over time [23, 12] and also make a long-term case for pursuing EA, however, the topic of measurement is discussed in the chapter 5.4 'Approaches to Value Measurement'.

5.3.4 Organizational Quality

The organizational context and culture are frequently argued to influence how effectively EA initiatives are embraced and embedded. Several studies suggest that directly linking EA efforts to business objectives and communicating the importance of EA across the enterprise may foster alignment and stakeholder buy-in [5, 39]. Similarly, the McKinsey report notes that active participation of architects in strategic discussions and the empowerment of stakeholders is believed to enhance the responsiveness and adaptability of EA practices [3].
It is also posited that the strategic positioning of EA within the organizational hierarchy is crucial, as when EA is supported by top-management commitment and adequate resources, this can serve as a critical enabler of EA success [3, 19, 25, 4]. Awareness-raising and securing stakeholder acceptance are said to reduce resistance and accelerate adoption, while addressing social and political dynamics is suggested to help establish a shared understanding of EA objectives [19, 25]. The Strategy& report further proposes that integrating EA into strategic planning, promoting it organization-wide, and embedding it within existing processes may ensure that EA benefits endure and evolve alongside the enterprise [23]. In addition, a well-structured business environment, which is characterized by cross-functional collaboration and competitive pressures, is argued to reinforce the value delivered by EA initiatives [12]. Finally, ongoing staff training and the development of internal expertise are often cited as mechanisms that foster organizational learning and maturity in EA capabilities [19, 23].

In summary, the literature suggests that the realization of EA benefits depends on the effective interplay of four key dimensions: product quality, infrastructure quality, service delivery quality, and organizational quality. Each of those contributes to successful EA outcomes when aligned with business objectives and supported by appropriate governance, tools, processes, and stakeholder engagement [14]. While empirical validation of these relationships remains limited, this framework offers a structured perspective for understanding how diverse factors collectively enable the translation of EA initiatives into tangible organizational value. For deeper insight, readers are referred to "A comprehensive EA benefit realization model – An exploratory study" by Lange et al. [19] and "Understanding the Benefits and Success Factors of Enterprise Architecture" by Jusuf & Kurnia [14] are suggested.

5.4 Value measurement

Based on the previous chapters it can be said that Enterprise Architecture gets credited with a wide range of benefits, promoting platform quality and cost reduction, supporting and guiding strategic decision or improving agility. However, in the daily operations of any enterprise, project initiatives demand clear, often monetary, evidence of success. It is therefore essential to define measurable, goal-aligned metrics that can be tracked and reported to demonstrate EA's tangible value. Even though the need for effective measurement in organizations is increasing, it still remains a relatively complex and difficult process to connect EA actions with organizational improvements [29].

5.4.1 Frameworks and Models

Traditionally, value is associated with a financial return on investment, and it has been argued that EA can be measured with the very same metrics. Using classical cost/income ratio is proposed by Strategy& to measure savings related to cost reduction by Enterprise Architecture, this supported by cost saving targets is said to quantify benefits related to cost reduction [23]. However, quantifying all benefits and cost can be very challenging, as EA often presents intangible costs and benefits. Linking the causality and impact of architecture would therefore require very deep financial knowledge and could even then not present a holistic value calculation [29]. One of the interviewees in the paper [36] states this problems with regards to the governance value of EA, where it can be used to validate the necessity of new applications based on the current landscape *"We stopped undesirable investment in assets that we know have been sunsetted. Great! Measure the value. What, dollar value? You know it is right. It is like saying—what is the value of speed tickets, speed cameras?"* [36]. This can be seen as an example where EA can lead to cost savings, but where the measurability is hindered.

This leads to the necessity to track also other metrics such as operational metrics and performance indicators. To name a few, [23] suggests time-to-market for new products which is suggested to use for measuring the agility and innovation benefits. According to [25] process and product quality measures can be used to track the quality benefits of EA. Furthermore, the papers also suggests before-after analysis when reviewing IT portfolios where the numbers of interfaces and technologies can be tracked [25]. The quality dimension around customer centricity can be collected via classical surveys [25]. [23] also suggest to combine this with basic metrics such as number of new channels and services developed for customers can be measured as well [23]. While more complex possibilities would be to use comparative metrics such as customer care performance changes for specific channels or services [23]. Another way of measuring success is described by banking executive, interviewed by Strategy& *"the percentage of funding allocated on the basis of architecture deliverables, value of the input into projects by the key stakeholders, how creative and proactive we are in our solution architectures, and our understanding of the business and its problems. Measures are scored through a survey sent to stakeholders."* [23].

Looking at those different approaches to measurement, it seems complex to assess the holistic value of 'doing' EA, one proposed solution to tackle this as a whole comes from Rodrigues and Amaral [29]. They state that an often approach is the Balanced Scorecard, however, according to them, this is still not an ideal approach which is why the propose a Value tree approach in their paper "Issues in Enterprise Architecture Value" [29]. A value tree is described as a method of systematically and visually connecting business operational metrics to shareholder value and financial metrics. This approach focuses on a small number of selected value drivers, that have a significant impact and can be constantly measured and controlled by the management an example is shown below.

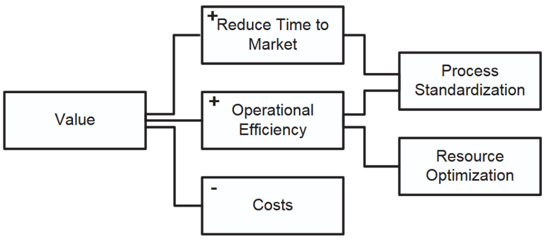

Figure 5.1: Value Tree Example [29]

To explain the terminology a little more, according to Rodriges and Amaral, value drivers are any variables that affect the value of the organization, on either cost or revenue side. To translate this onto Enterprise Architecture, value drivers mean any variable that therefore provides benefit or cost of financial or other matter to the organization. The most pressing concerns when constructing the value tree is the selection of the key value drivers, as stated above, Rodrigues et al. suggest choosing those drivers that have the most impact, but are also decomposable into smaller components. This is done to enable the creation of different levels of value drivers, based on their abstraction. The tree therefore consists of low-level, specific drivers that influence and drive high-level abstract drivers. This is suggested to enable a higher level of detail, furthermore, those levels can be also understood the drivers of different stakeholders. As more low-level drivers can be associated with more operative work, while more abstract values are tighter connected to strategic positions. Another important factor that [29] identified was that the value depends on both on long- and short term value of the organization, to cover both impact areas, but more to that in the next subchapter. To finish the value tree, it is necessary to define and estimate corresponding metrics for different parts of the tree.

5.4.2 Connection to timeline

One of the reasons that make the measurement of Enterprise Architecture Value additionally challenging is the time perspective of EA benefits. As stated by [29], economic pressure leads to companies searching quick results from their investments and efforts. However, effects of Enterprise Architecture may not always be tangible in such time horizons. To further investigate this, the paper “Stakeholders Perspectives on time horizon and quantification of Enterprise Architectures benefits/value drivers” [30] presents the results of a survey sent out to different EA experts that questioned the measurability of EA benefits and whether those are realized short- or long-term. Based on a list of 29 Benefits, ranging from ‘(improved) Alignment’ to ‘(reduced) Complexity’, those two dimensions were interrogated based on the questions, whether those benefits can be realized in the short term, specifically less than a year and whether or not those benefits can be measured in any way. Interestingly, out of the of 29 benefits only 4 of them were considered to be short-term from the majority of experts. Furthermore, 14 of the benefits were categorized as long-term by 85% or more of the experts, which shows a strong tendency towards long-term realization. The second question that was put, was about measurability and here, the overall tendency is even stronger. Out of 29 value claims, only two were considered to be measurable. With only ‘(reduced) Costs’ being categorized as measurable with a strong tendency of the experts. When connecting those two questions, it becomes clear, that none of the short-term values is categorized as measurable, thereby complicating the delivery of quick results from EA [30].

Bringing this together means that there seems to be a relevant challenge in supporting benefit claims with clear and objectively measured metrics. Even though classical financial metrics are partially applicable to quantify positive effects of Enterprise Architecture, traditional systems seem to be not able to capture intangible long-term benefits. Hybrid models of traditional metrics and more open approaches seem to present a potential solution, e.g. the value tree concept proposed by [29].

5.5 Implications for Interview-Based Validation

As this work is not just aimed at analyzing literature, but also at providing insights and guidance for potential empirical studies, this section presents derived implications based on the conducted research. when looking at empirical interviews regarding the actual value of ‘doing’ Enterprise, there can be several implication drawn from our work.

5.5.1 Selection of Interview participants

During this research process it became clear, that Enterprise Architecture and therefore, the 'doing', thereby actively creating Enterprise Architecture and planning a 'to-be' state involves multiple different stakeholders. As this creation involves such different positions and personas, so do the benefits. while some benefits may cover a more abstract view of Enterprises such as values around strategy and innovation, other, such as IT quality and standardization take place on an implementation level. We have seen in our research, that literature that mainly focuses on portraying the viewpoint of executives often fail to cover benefits that happen more on the implementation level. If one would like to research organizational benefits of 'Doing' EA, practitioners from different fields, departments and seniority level are helpful to deliver a broad and realistic picture. Below, there is a proposed selection of relevant stakeholders that could be included in such interviews:

- **Enterprise Architects**: Practice leads with a holistic "to-be" perspective and key in regulative, instructive, and informative roles of EA.
- **IT & Solution Architects**: Responsible for translating EA principles into concrete implementations.
- **Business Stakeholders**: Recipients of EA-driven governance and alignment processes.
- **Project Managers**: Experience EA's impact on cost, risk, and complexity in delivery.
- **Senior Executives (CIO, CTO, COO)**: Speak to high-level strategic and financial ROI expectations.

Including different viewpoints might let one see lots of benefits while still noticing each group's own concerns. Getting stakeholders involved can further more strengthen their feeling of being recognized and more willing to support the results. This also applies to varying the industry backgrounds based on the research scope, as the maturity of EA might highly differ across market segments.

5.5.2 Content of Interviews

Based on our conducted research, we propose a semi-structured approach, centered around the main dimensions of Enterprise Architecture value that were identified in this paper. Therefore, we suggest questioning about benefits in the following dimensions:

- *Cost & Time Reduction*
- *Complexity Management*
- *Risk Management*
- *Quality increase*
- *Strategy & Innovation*
- *Decision Making & Agility*
- *Alignment & Governance*

Exemplary questions for the respective categories could be *"How has EA influenced* *complexity management in your initiatives?"*, this approach is proposed as it combines a rough direction without pushing participants to far into one singular direction, this can also be achieved by asking for specific examples for ways in which EA has contributed to the mentioned dimensions. Furthermore, a lot of literature such as [4] suggests, that EA adaption and effectiveness highly depends on organizational circumstances and the environment. To put this into the equation, initial questions about the organizational context would be necessary. Those questions could cover topics like overall digital maturity of the organization, maturity of EA or positioning of Enterprise Architecture in the organization(decentralized, centralized etc.). Those insights further contextualize the overall priority and maturity of EA in the given enterprise. Given the ambiguous understanding of EA value measurement, it could also be interesting wether or not there exist certain metrics to quantify EA value in the given organization.

5.5.3 Methodological Considerations

Next to the overall content, there are also a few additional things to look out for. As the topic of this work is rather 'doing' Enterprise Architecture than 'documenting' EA, this distinction is important to consider as the different activities in EA often merge seamlessly into one another. Therefore, it could be helpful to focus on transformative projects and initiatives and specifically ask about such occurrences to stay in scope. As already mentioned before, contextual questions can help in understanding the organizational environment. Factors such as industry, company size, regulatory environment can heavily impact the value-add of Enterprise Architecture. Furthermore, it might also occur that surveyed participants may cite best-practice claims without concrete experience, as those claims might be often told to respective positions without them having them actually experienced. This can be mitigated with counter checking by asking for concrete projects. Anonymity is of course a basis for those interviews as such a research should not be supposed to play off business units against each other.

CHAPTER 6

Conclusion

This thesis aimed to explore the value claims associated with 'doing' Enterprise Architecture, meaning the active creation and modeling of to-be states of Enterprises. To do this, academic and industry literature was analyzed and systematically reviewed. By developing a categorization framework, this thesis identified, synthesized and compared benefit dimensions of EA across two primary perspectives: IT-related and business-related. Thereby this thesis tried to contribute to a more nuanced understanding of Enterprise Architecture's value proposition.

The analysis revealed that the value of doing EA is multifaceted and emerges from the interconnection of its informative, instructive, and regulative roles. IT-related benefits include cost and complexity reduction, risk mitigation, and quality improvement, often supported through enhanced transparency, standardization, and architectural reuse. On the business side, EA supports strategic alignment, informed decision-making, organizational agility, and innovation enablement. Importantly, these value claims are not isolated but form a network of mutually reinforcing mechanisms that support enterprise transformation. The thesis also highlighted differences in how academic and industry sources construct and argue these value claims, which showed differences that reflect distinct objectives and rhetorical strategies.

However, the research also showed ongoing challenges related to the realization and measurement of EA benefits. Despite clear consensus on the potential value of EA, both scholars and practitioners struggle to quantify this value in practice. Long-term benefit horizons, intangible outcomes, and limited causal traceability complicate efforts to establish objective value attribution. While frameworks such as the value tree model propose more nuanced approaches, there remains a gap between conceptual value understanding and operational measurement in organizations.

The findings of this study should also be interpreted in light of several limitations. First, although a effort was made to include high-quality industry sources, these documents are not peer-reviewed and often reflect commercial objectives, which may introduce bias. Second, a substantial portion of the academic literature used in this thesis consists of secondary studies and reviews, rather than empirical primary research. As such, the benefit claims analyzed here are often second-order interpretations rather than direct observations. These limitations emphasize the need for cautious interpretation and further empirical validation of the results.

Given these findings, future research should aim to validate the identified value claims through interviews with stakeholders involved in EA initiatives. Particular attention should be paid to how benefits are experienced across roles and organizational contexts, and how different maturity levels of EA capabilities affect benefit realization. By further supporting the value of doing EA in practice, such research can help bridge the gap between theoretical claims and organizational impact, ultimately supporting more informed decision-making about when, why, and how to invest in EA.

Overview of Generative AI Tools Used

The following generative AI tools were used to support the thesis writing process. Their use was strictly limited to auxiliary tasks such as enhancing clarity, correcting grammar, and rephrasing unclear or overly complex sentences. At no point did these tools replace the author's own intellectual contributions. All content generated with the assistance of language models was critically reviewed, revised, and verified by the author to ensure compliance with academic standards and integrity. The use of generative AI served solely as a writing aid, enabling consistent, precise, and accessible scholarly expression.

- **DeepL Translate:** Translation of individual sentences from German to English.
- **ChatGPT (GPT-4 o4-mini-high):** Rephrasing sentences and summarizing selected academic papers.
- **Perplexity.ai:** Support in identifying relevant publications during initial literature search.

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